



What is artificial intelligence (AI) in networking?

Artificial intelligence (AI) is a field of study that gives computers human-like intelligence when performing a task. When applied to complex IT operations, AI assists with making better, faster decisions and enabling process automation.

Explore what AI-enhanced infrastructure can do for your organization.

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[Artificial intelligence](#) simulates intelligent decision making in computers. It's not uncommon for some to confuse artificial intelligence with machine learning (ML) which is one of the most important categories of AI. Machine learning can be described as the ability to continuously "statistically learn" from data without explicit programming.

What are the advantages of AI and ML in networking?

The benefits of implementing AI/ML technology in networks are becoming increasingly evident as networks become more complex and distributed. AI/ML improves troubleshooting, quickens issue resolution, and provides remediation guidance. It brings about critical insights to improve user and application experience. AI/ML can be used to respond to problems in real-time, as well as predict problems before they occur. It also augments security insights by improving threat response and mitigation.

How does AI transform networking?

Using AI and ML, network analytics customizes the network baseline for alerts, reducing noise and false positives while enabling IT teams to accurately identify issues, trends, anomalies, and root causes. AI/ML techniques, along with crowdsourced data, are also used to reduce unknowns and improve the level of certainty in decision making.



How important are AI and ML technologies in managing and monitoring today's networks?

Collecting anonymous telemetry data across thousands of networks provides learnings that can be applied to individual networks. Every network is unique, but AI techniques let us find where there are similar issues and events and guide remediation. In some cases, machine learning algorithms may strictly focus on a given network. In other use cases, the algorithm may be trained across a broad set of anonymous datasets, leveraging even more data.

AI/ML technology

Network automation and AI/ML

IT can gain insights through analytics and AI/ML that guide more trusted automation processes that lower the cost of network operations and provide users with an optimal connected experience. These technologies help IT automate:

- The deployment and management of network policies
- The integration of zero-trust security solutions to help ensure network consistency
- The identification and classification of devices on the network

Over time, AI will increasingly enable networks to continually learn, self-optimize, and even predict and rectify service degradations before they occur.

AI and ML models



Using sophisticated AI/ML models, users can view network-health benchmarks that are based on network data that is collected over time. Insights for network optimization include:

- Long-term variations in performance trends
- Comparison of network health within an enterprise or with industry peers

AI/ML and telemetry

Through a network controller and management dashboard, telemetry data from the network can be ingested and processed through AI/ML engines to:

- Identify anomalies
- Eliminate false positives
- Suggest remediation actions

AI and machine reasoning (MR)

Machine reasoning (MR) is another important category of AI. Machine reasoning uses acquired knowledge to navigate through a series of possible options toward an optimal outcome. MR is well suited for solving problems that require deep domain expertise. Humans need to explicitly capture all the knowledge a priori in order for a machine reasoner to be able to operate on new data. MR is a wonderful complement to ML because it can build on the conclusions presented by ML and analyze possible causes and potential improvement options.

AI/ML and predictive analytics

Simply put, predictive analytics refers to the use of ML to anticipate events of interest such as failures or performance issues, thanks to the use of a model trained with historical data. Mid- and long-term prediction approaches allow the system to model the network to determine where and when



actions should be taken to prevent network degradations or outages from occurring.

AI-enabled networking use cases

AI/ML for improving Wi-Fi performance

Using machine learning, NetOps teams can be forewarned of increases in Wi-Fi interference, network congestion, and office traffic loads. By learning how a series of events are correlated to one another, system-generated insights can help foresee future events before they happen and alert IT staff with suggestions for corrective actions.

AI/ML for tracking IoT endpoints

AI/ML is beneficial for Internet of Things (IoT) deployments. IoT devices can have a broad set of uses and can be difficult to identify and categorize. Machine learning methods can be used to discover IoT endpoints by using network probes or using application layer discovery techniques.

Machine learning for policy automation

Machine learning can be used to analyze traffic flows from endpoint groups and provide granular details such as source and destination, service, protocol, and port numbers. These traffic insights can be used to define policies to either permit or deny interactions between different groups of devices, users, and applications.

Machine reasoning for improved licensing

Machine reasoning can parse through thousands of network devices to verify that all devices have the latest software image and look for potential vulnerabilities in device configuration. If an operations team is not taking advantage of the latest upgrade features, it can flag suggestions.



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